

University of Hohenheim

**Time Series Approaches to Cocoa Price
Risk**

Master Thesis

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1 Introduction

Volatility forecasting is fundamental to financial risk management, portfolio allocation, and derivative pricing (Poon and Granger 2003). Two modeling paradigms dominate the literature. The first treats volatility as latent and inferred from returns; the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model of Bollerslev (1986) is the canonical approach. The second, enabled by high-frequency data, treats volatility as directly observable through realized measures, and the Heterogeneous Autoregressive (HAR) model of Corsi (2009) has become the dominant forecasting framework within this paradigm, capturing long-memory properties of volatility in a parsimonious, OLS-estimable structure.

The majority of volatility forecasting research focuses on conditional mean forecasts: the expected level of future volatility given current information. Yet the conditional mean provides an incomplete characterization of future volatility. When the volatility distribution is skewed or fat-tailed—as is typical for commodity markets—the conditional quantiles can diverge substantially from the mean, and understanding how the entire distribution shifts with predictor values becomes important. Quantile regression, introduced by Koenker and Bassett (1978), provides a natural framework for estimating conditional quantiles directly, without distributional assumptions.

Cocoa futures present a particularly relevant case for distributional volatility forecasting. Production is geographically concentrated in West Africa, biologically slow to adjust, and exposed to both disease and climate shocks. Between 2022 and 2024, international cocoa prices approximately tripled—an increase unprecedented in the commodity’s modern trading history. Despite this salience, dedicated cocoa volatility research remains limited, and the application of HAR-based quantile regression to cocoa—or to any agricultural commodity—is an open gap in the literature. At long forecast horizons, where the autoregressive signal from lagged volatility fades, climate variables offer an additional source of predictive information. The El Niño–Southern Oscillation (ENSO) influences West African rainfall patterns with effects on cocoa yields materializing 6–12 months after the climate signal, and recent evidence shows that ENSO improves agricultural commodity volatility forecasts at precisely these longer horizons (Bouri et al. 2021; Bonato et al. 2023).

This thesis addresses two research questions:

RQ1 Can HAR-based quantile regression models forecast cocoa futures volatility better than baseline models (historical volatility, GARCH, HAR-OLS) at horizons from 1 day to 12 months?

RQ2 Does the ENSO climate index improve volatility forecasts, especially at long horizons?

We address these questions by estimating HAR-QR models at multiple quantiles ($\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$) and horizons (1 day, 1 week, 1 month, 3 months, 6 months, 12 months). At each horizon, we compare out-of-sample forecast accuracy against standard benchmarks: historical rolling volatility, GARCH(1,1), and HAR estimated by ordinary least squares. For long horizons, we test whether adding ENSO improves forecasts, with particular focus on the upper quantile ($\tau = 0.95$).

The thesis makes three contributions. First, it provides the first application of HAR-QR to an agricultural commodity, extending a framework previously confined to equity and precious-metals markets. Second, it tests HAR-QR at horizons of 6–12 months—far beyond the current literature limit of approximately one month (Lyócsa and Molnár 2021). Third, it tests whether ENSO enhances quantile volatility forecasts for cocoa, a combination untested in the literature.

The remainder of the thesis is organized as follows. Section 2 reviews the literature on volatility measurement, HAR models, quantile regression in finance, and climate-commodity linkages, establishing the research gap this thesis addresses. Section 3 specifies the HAR-QR model, the multi-horizon projection approach, and the evaluation framework. Section 4 describes the ICE London cocoa futures data and validates the daily volatility proxy against realized volatility from intraday data. Section 5 presents empirical results organized by forecast horizon. Section ?? interprets findings, discusses limitations, and considers practical implications.

2 Literature Review

This section reviews the literature underlying volatility forecasting for cocoa futures. We begin with the cocoa market itself, then trace how volatility measurement evolved from simple estimators to the HAR framework. We then review quantile regression for volatility and ENSO as a climate predictor at long horizons, followed by the evaluation methods used to compare forecasts. The intersection of these strands identifies the gap this thesis addresses: no study applies HAR-based quantile regression to any agricultural commodity, and no HAR-QR study extends beyond approximately one month.

2.1 The Cocoa Market

Cocoa occupies an unusual position among agricultural commodities. Its production is geographically concentrated, biologically slow to adjust, and structurally exposed to climate variability in ways that set it apart from grains or oilseeds. These features interact to produce a market where supply disruptions are both frequent and slow to resolve—precisely the conditions under which distributional, rather than mean, volatil-

ity forecasts become most valuable.

The West African cocoa belt accounts for approximately 70% of world production, with Côte d’Ivoire and Ghana alone responsible for more than half of global output (Schroth et al. 2016; Läderach et al. 2013). Unlike grains or oilseeds, whose production is spread across multiple continents, cocoa’s concentration means that disruptions in a single region can move global prices substantially. Production is overwhelmingly smallholder-based, with approximately 90% grown on farms smaller than five hectares (Bensch et al. 2022). Rogna and Tillie (2025) identify the stock-to-grind ratio as the primary long-run price determinant, finding a significant cointegrating relationship between prices and inventory levels. Since the mid-2000s, growing financial participation has added a further source of volatility (Bensch et al. 2022; Gilbert and Morgan 2010).

A critical structural feature is the cocoa tree’s biological rigidity. New trees require five to seven years before reaching full productive capacity, with an economic lifespan of 30 to 40 years (Asante et al. 2025). Unlike annual crops, cocoa supply responds to market signals slowly, and the resulting low short-run supply elasticity amplifies price volatility whenever demand or supply shocks occur.

Cocoa is also among the most climate-sensitive perennial crops. West Africa is characterized by “some of the most variable climates on the planet” (Läderach et al. 2013), and drought during the critical September–December pod development period reduces yields substantially. The El Niño–Southern Oscillation (ENSO) is the primary near-term mechanism through which climate variability affects cocoa supply; Section 2.5 discusses this channel and its implications for long-horizon volatility forecasting.

The most striking recent illustration of these compounding dynamics is the 2022–2024 price crisis, during which international cocoa prices approximately tripled (Rogna and Tillie 2025). The crisis was driven by the convergence of the Cocoa Swollen Shoot Virus Disease (CSSVD) in Ghana and adverse El Niño–related drought across Côte d’Ivoire.

Despite the salience of cocoa in global commodity markets, the volatility forecasting literature has paid it limited attention. Bensch et al. (2022) provide the most comprehensive study of cocoa price determinants, finding that fundamentals are quickly integrated into prices, but do not extend to multi-horizon forecasting or tail-risk quantification. Bergander (2025) apply LSTM neural networks with weather data but find no improvement in point forecasts, plausibly because markets anticipate weather shocks before physical effects materialize. Within broader commodity surveys, Alfeus and Nikitopoulos (2022) include cocoa among 22 markets and find that monthly volatility is the most informative component—evidence of slow-moving, structural dynamics—but offer no focused analysis. No study applies HAR-based models specifically to cocoa volatility, and none has examined quantile-based tail-risk forecasting for this market.

2.2 The Evolution of Volatility Measurement

A foundational challenge in volatility research is that volatility is latent: unlike prices, which are directly observable, it must be estimated from price movements (Poon and Granger 2003). Every volatility measure is an approximation, and the choice of estimator shapes both forecast quality and evaluation validity (Patton 2011). The measurement literature has evolved through three successive approaches, each reducing the statistical noise of the volatility proxy.

The earliest volatility estimates used only closing prices. The random walk model, exponential smoothing, and rolling-window standard deviations all extrapolate past variation into the future (Poon and Granger 2003). These methods are simple and require minimal data, but the squared close-to-close return is a highly imprecise estimator of daily variance. Nonetheless, Poon and Granger (2003) review 93 studies and find that “for forecast horizons that are longer than 6 months, a simple historical method using low frequency data over a period at least as long as the forecast horizon works best”—a finding that motivates the inclusion of rolling historical volatility as a benchmark in this thesis.

A significant efficiency improvement came from exploiting the full daily price range. Parkinson (1980) demonstrated that the high-low range achieves efficiency gains of approximately five times over close-to-close methods. Rogers and Satchell (1991) extended this to allow for non-zero drift. However, both estimators assume continuous trading—an assumption violated by commodity futures markets, which experience regular overnight closures.

Yang and Zhang (2000) synthesized these approaches into an estimator that is unbiased, drift-independent, and robust to opening jumps. By estimating overnight and trading-session variance separately and combining them optimally, the Yang-Zhang estimator achieves efficiency gains of 7–8 times over close-to-close methods. For commodity markets, where overnight information frequently causes opening gaps, this robustness is particularly valuable.

The availability of intraday transaction data in the late 1990s enabled a fundamentally different approach. Andersen and Bollerslev (1998) and Andersen et al. (2001) demonstrated that realized volatility—the sum of squared intraday returns—provides a far more precise measure of daily variance, resolving much of the apparent poor performance of earlier models. However, high-frequency data are plagued by microstructure noise and are often unavailable for commodity markets over long samples. Clements et al. (2024) provide evidence that HAR models applied to daily OHLC-based volatility achieve comparable forecasting performance at medium and long horizons, validating the use of daily data when intraday records are limited—as is the case for cocoa futures over multi-decade samples.

These developments establish the Yang-Zhang estimator applied to daily OHLC data as the volatility proxy for this thesis. The following section turns from the question of what to measure to the question of how to model and forecast its temporal dynamics.

2.3 The Heterogeneous Autoregressive Model

Two modeling frameworks are central to this thesis: GARCH, which serves as the main benchmark, and the Heterogeneous Autoregressive (HAR) model, which is the primary forecasting framework. Both capture a defining feature of volatility that range-based estimators ignore: its tendency to cluster over time.

Volatility clustering was first formalized by Engle (1982), and Bollerslev (1986) generalized this into the GARCH framework, capturing both clustering and mean reversion. The GARCH(1,1) has proven remarkably difficult to beat: Hansen and Lunde (2005) evaluate 330 volatility models and conclude that “nothing beats a GARCH(1,1),” establishing it as the natural benchmark. However, GARCH cannot replicate the slow, hyperbolic decay of volatility autocorrelations observed empirically—a property known as long memory. As Corsi (2009) observes, “standard GARCH and stochastic volatility short-memory models appear like white noise once aggregated over longer time periods.”

The HAR model, introduced by Corsi (2009), addresses this limitation through a parsimonious structure that captures long memory without the estimation difficulties of fractionally integrated models. It builds on the *Heterogeneous Market Hypothesis* of Müller et al. (1997), who documented that coarse-grained volatility (e.g., weekly) predicts fine-grained volatility (e.g., daily) more strongly than the reverse. This asymmetry reflects the presence of traders with different investment horizons—intraday speculators, swing traders, and long-term institutional investors—whose actions generate volatility that cascades across frequencies. Corsi formalized this insight into an additive model with daily, weekly, and monthly volatility components: a restricted AR(22) estimable by OLS that reproduces the autocorrelation patterns of long-memory processes.

Since its introduction, the HAR framework has been extended with jump components, realized semivariance, and time-varying coefficients (Andersen et al. 2007; Patton and Sheppard 2015; Tian et al. 2017). However, for agricultural commodities, Degiannakis et al. (2022) find that “sophisticated HAR-type models are not capable of outperforming the simple HAR in an out-of-sample exercise,” suggesting that parsimony is valuable and additional complexity must be justified by clear forecast improvements.

For cocoa specifically, Alfeus and Nikitopoulos (2022) include it among 22 commodities in a broad HAR study, finding that “monthly volatility matters the most” for

cocoa—evidence that longer-term participants exert particular influence on commodity volatility dynamics. Haugom et al. (2016) provide complementary evidence that “the effect from the monthly volatility component is strongest for all assets when predicting conditional tails.” Despite this, no study focuses specifically on cocoa volatility dynamics or applies HAR to horizons beyond one month.

2.4 Quantile Regression for Volatility

Standard volatility models—whether GARCH or HAR—estimate the conditional mean of volatility. Yet when the underlying process exhibits fat tails and regime shifts, the conditional quantiles can diverge substantially from the mean. Quantile regression, introduced by Koenker and Bassett (1978), estimates conditional quantiles directly, with each quantile having its own parameter vector. This allows the relationship between predictors and the dependent variable to differ across the distribution without imposing parametric assumptions on the error distribution.

Baur and Dimpfl (2019) demonstrate that quantile autoregression encompasses both the conditional mean and conditional variance of the return distribution through the inter-quantile range. Their approach is robust to jumps and produces variance forecasts comparable to, and in some cases more accurate than, GARCH and HAR alternatives. Haugom et al. (2016) introduced HAR-QREG, combining the multi-horizon HAR structure with quantile regression, and show that the model passes conditional coverage tests at substantially higher rates than historical simulation.

HAR-QR has since been applied in equity and precious-metals markets. Lyócsa and Molnár (2021) develop a complete subset quantile regression variant achieving significant improvements at horizons up to 22 trading days; Huang et al. (2022) confirm the model’s superiority using high-frequency data for Chinese indices; and Song et al. (2025) show that HAR-QR parameters improve machine learning forecasts for gold.

Despite this growing body of work, two gaps remain. First, no study applies HAR-QR to any agricultural commodity (Degiannakis et al. 2022; Bonato et al. 2023). Second, existing studies are confined to short horizons—Lyócsa and Molnár (2021) extend to 22 trading days, the longest in the literature. Whether quantile forecasts retain predictive value at horizons of 6–12 months remains untested.

2.5 Climate Variables and Commodity Volatility

At long forecast horizons, the autoregressive signal from lagged volatility fades. Climate variables offer a potential source of predictive information that operates on precisely these longer timescales.

The El Niño–Southern Oscillation (ENSO) is an irregularly periodic fluctuation in Pacific sea surface temperatures that affects weather patterns globally, cycling between

warming (El Niño) and cooling (La Niña) phases every two to seven years. For cocoa, the transmission mechanism is direct: during El Niño phases, West Africa typically receives below-normal rainfall during the growing season, reducing yields in subsequent harvests. Schroth et al. (2016) document that “drought years regularly affect cocoa yields in West Africa and have particularly done so during severe El Niño years.” With the main crop harvested from October through March and the mid-crop from May through August, ENSO-driven rainfall anomalies translate into production changes six to twelve months after the climate signal—a transmission lag that aligns with the long forecast horizons this thesis examines. Importantly, markets respond not only to realized weather but to climate expectations: Bergander (2025) observe that roughly 85% of the eventual price impact of a drought shock is reflected in futures prices before the actual crop loss occurs.

At the commodity price level, Ubilava (2018) examines ENSO–price relationships across 43 commodities, finding significant effects for tropically grown agricultural products including cocoa, with “more amplified price responses during El Niño events.” Recent work has extended this analysis from price levels to volatility. Bouri et al. (2021) find that ENSO improves oil volatility forecasts at horizons of 2–4 years—far beyond where lagged volatility alone provides value. Bonato et al. (2023) extend this to 16 agricultural commodities, finding that “the evidence of predictive value of El Niño and La Niña events strengthens at the longer term forecast horizons”—precisely where standard autoregressive models lose power. More broadly, Guo et al. (2025) demonstrate that climate risk indices significantly improve commodity volatility forecasts, with models incorporating climate risk “outperforming traditional ones in economic value.”

These findings establish ENSO as a physically motivated, exogenous predictor with demonstrated value for commodity volatility at long horizons—but this combination has not been tested for cocoa, nor has it been combined with quantile regression.

2.6 Evaluating Volatility Forecasts

A distinctive challenge in evaluating volatility forecasts is that the variable of interest is unobservable, even ex post (Patton 2011). Every volatility measure is an imperfect proxy, and noise in the proxy can produce misleading model rankings depending on the loss function used.

Patton (2011) derives conditions for loss functions that yield rankings invariant to the choice of volatility proxy. Among common loss functions, only mean squared error (MSE) and QLIKE satisfy these conditions. Non-robust loss functions can produce rankings that deviate by factors of one quarter to three, explaining widespread conflicting findings across studies. Between the two robust measures, QLIKE penalizes

under-prediction proportionally more heavily, making it suitable when the forecasting goal involves risk assessment (Lyócsa and Molnár 2021; Clements et al. 2024; Ghysels and Marcellino 2018).

For pairwise model comparison, the Diebold-Mariano-West test (Diebold and Mariano 1995) tests for equal predictive accuracy and is robust to serial correlation in forecast errors. When multiple models are compared simultaneously, Hansen et al. (2011) propose the Model Confidence Set (MCS), which identifies the set of models $\mathcal{M}^*$ containing the best model at a given confidence level. The MCS “acknowledges the limitations of the data”: informative samples yield small confidence sets, while uninformative samples include many models—particularly relevant when evaluation noise from an unobservable target reduces the power to discriminate (Clements et al. 2024; Degiannakis et al. 2022).

For quantile forecasts, Christoffersen (1998) establishes conditional coverage testing: a well-specified quantile forecast at level τ should produce violations at rate τ , and these violations should be serially independent. The quantile loss function—the check function used in estimation—also serves as a natural evaluation metric, penalizing over- and under-prediction asymmetrically according to τ (Lyócsa and Molnár 2021). Puke and Schweikert (2026) show that aligning the estimation and evaluation loss function produces significant forecast gains; since HAR-QR uses the check function for both estimation and evaluation, it is inherently coherent by construction.

3 Methodology

This section specifies the econometric framework for forecasting cocoa futures volatility at horizons from one day to twelve months. We first define the volatility measure, then specify the HAR-QR model and its ENSO-augmented extension, describe the benchmark models, and finally lay out the evaluation framework. Table 1 provides an at-a-glance summary of the five models before the detailed specifications follow.

3.1 Volatility Measure

We measure daily volatility using the Yang-Zhang estimator, which combines overnight and trading-session variance components:

$$\hat{\sigma}_{YZ}^2 = \hat{\sigma}_O^2 + k\hat{\sigma}_C^2 + (1 - k)\hat{\sigma}_{RS}^2, \quad (1)$$

where $\hat{\sigma}_O^2$ captures overnight variance, $\hat{\sigma}_C^2$ captures open-to-close variance, and $\hat{\sigma}_{RS}^2$ is the Rogers-Satchell component that exploits the daily range. The weighting factor $k = 0.34/(1.34 + (n + 1)/(n - 1))$ minimizes estimator variance.

Table 1: Overview of models compared in this study, ordered from simplest to most complex.

Feature	Benchmarks			Main models	
	Hist. Vol.	GARCH(1,1)	HAR-OLS	HAR-QR	HAR-X-QR
Model class	Non-param.	ARCH-type	Linear reg.	Quantile reg.	Quantile reg.
HAR lags ($d/w/m$)	No	No	Yes	Yes	Yes
ENSO regressor	No	No	No	No	Yes
Quantile output	Empirical	Simulation	None	Direct	Direct
Estimation	—	MLE	OLS	Check fn.	Check fn.
Multi-period design	Rolling window	Recursive	Direct	Direct	Direct
Thesis role	Naive BM	Param. BM	HAR benchmark	Main (RQ1)	Main (RQ2)

Note: Direct = horizon-specific model estimated by direct projection; Recursive = closed-form GARCH mean recursion, quantiles via Monte Carlo simulation; Check fn. = check-function (quantile) loss; BM = benchmark; Hist. Vol. = rolling 252-day window with empirical quantiles.

This choice is motivated by two considerations. First, Yang-Zhang achieves efficiency gains of 7–8 times over close-to-close estimators, making it well-suited to commodity futures where overnight gaps are common. Second, it can be computed over our full 40-year sample, unlike realized volatility which requires high-frequency data available only since 2019. We validate the Yang-Zhang proxy against five-minute realized volatility in Section 4, comparing rolling-window aggregates (22-day averages) of both measures via a Mincer-Zarnowitz regression. This comparison at the multi-day level aligns with the HAR model’s use of averaged volatility as both predictors and targets.

3.2 The HAR-QR Model

Combining the HAR framework with quantile regression yields the HAR-QR model. The HAR-QR model for horizon h and quantile τ is

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)}, \quad (2)$$

where $\mathcal{F}_t$ denotes the information set at time t . The volatility components capture variation at three frequencies:

$$\sigma_t^{(d)} = \sigma_t, \quad (3)$$

$$\sigma_t^{(w)} = \frac{1}{5} \sum_{i=0}^4 \sigma_{t-i}, \quad (4)$$

$$\sigma_t^{(m)} = \frac{1}{22} \sum_{i=0}^{21} \sigma_{t-i}. \quad (5)$$

The key feature of quantile regression is that each quantile τ has its own coefficient

vector, allowing the relationship between predictors and future volatility to differ across the distribution. For the median ($\tau = 0.50$), the coefficients may resemble those from OLS estimation. For the upper quantile ($\tau = 0.95$), the coefficients capture how the predictors relate to volatility during high-volatility episodes. We estimate five quantiles: $\tau \in \{0.05, 0.25, 0.50, 0.75, 0.95\}$. The extreme quantiles ($\tau = 0.05$ and $\tau = 0.95$) characterize the tails of the conditional distribution, while the intermediate quantiles provide a fuller picture of how the distribution shifts with predictor values. The upper quantile $\tau = 0.95$ is of particular interest, as it captures the conditional upper bound of volatility—the regime where mean forecasts are most likely to understate actual outcomes.

The dependent variable is the average daily volatility over the forecast horizon, following the HAR literature convention (Ghysels and Marcellino 2018; Clements et al. 2024):

$$\bar{\sigma}_{t:t+h} = \frac{1}{h} \sum_{i=1}^h \sigma_{t+i}. \quad (6)$$

This averaging ensures comparability across horizons and aligns with the HAR model’s construction, where weekly and monthly components are themselves averages of daily volatility.

We adopt the *direct* forecasting approach: a separate model is estimated for each horizon h , using $\bar{\sigma}_{t:t+h}$ as the dependent variable. This contrasts with *iterated* forecasting, which estimates a one-step model and iterates forward. Ghysels and Marcellino (2018) find that direct forecasting is preferred for realized volatility models. For quantile regression, direct estimation is particularly natural, as it produces the conditional quantile at horizon h without the distributional assumptions required to iterate quantile forecasts. Our longest horizons (6–12 months) would require iterating hundreds of daily steps, amplifying any model misspecification.

We estimate HAR-QR at six horizons: $h \in \{1, 5, 22, 66, 132, 264\}$ trading days, corresponding to 1 day, 1 week, 1 month, 3 months, 6 months, and 12 months. The short horizons (1 day to 1 month) serve to validate the methodology against existing literature: HAR-QR has been tested at horizons up to 22 trading days (Lyócsa and Molnár 2021). The long horizons (6 and 12 months) constitute the core contribution, testing whether HAR-QR can produce useful forecasts at horizons where no prior study has ventured.

At long horizons, the autoregressive signal from lagged volatility fades. We therefore augment the HAR-QR model with the ENSO climate index:

$$Q_\tau(\bar{\sigma}_{t:t+h} \mid \mathcal{F}_t) = \beta_0(\tau) + \beta_d(\tau)\sigma_t^{(d)} + \beta_w(\tau)\sigma_t^{(w)} + \beta_m(\tau)\sigma_t^{(m)} + \gamma(\tau)\text{ENSO}_t, \quad (7)$$

where ENSO_t is the Niño 3.4 sea surface temperature anomaly index from NOAA, mea-

sured monthly. The coefficient $\gamma(\tau)$ captures the marginal effect of ENSO conditions on future volatility at quantile τ .

We use *contemporary* ENSO—that is, ENSO_t predicts $\bar{\sigma}_{t:t+h}$ —rather than explicitly lagged ENSO values. This specification is standard in the climate-commodity literature (Bouri et al. 2021; Bonato et al. 2023). The key insight is that the forecast horizon itself provides the transmission lag: when ENSO_t predicts $\bar{\sigma}_{t:t+264}$ (12 months ahead), there are 12 months for the climate-to-production-to-price transmission to unfold.

The transmission mechanism for cocoa operates through West African rainfall. El Niño phases typically bring drought conditions during the critical pod development period (September–December), reducing yields in the subsequent harvest (Ubilava 2018). With two harvest seasons per year (main crop October–March, mid-crop May–August), ENSO effects materialize 6–12 months after the climate signal, aligning with our forecast horizons. ENSO is exogenous to the cocoa market—Pacific Ocean temperatures cannot be caused by cocoa prices—and is publicly available with minimal delay.

We obtain monthly Niño 3.4 sea surface temperatures from NOAA’s Extended Reconstructed SST (ERSSTv5) dataset and convert them to anomalies by subtracting the 1991–2020 climatological monthly mean, following standard practice (Bouri et al. 2021; Bonato et al. 2023). Positive anomalies indicate El Niño conditions; negative anomalies indicate La Niña. Since the index is monthly, each trading day inherits the Niño 3.4 anomaly of its calendar month.

The HAR-QR model is estimated by minimizing the check function loss:

$$\hat{\beta}(\tau) = \arg \min_{\beta} \sum_{t=1}^{T-h} \rho_{\tau}(\bar{\sigma}_{t:t+h} - X_t' \beta), \quad (8)$$

where $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)})'$ for the basic model, or $X_t = (1, \sigma_t^{(d)}, \sigma_t^{(w)}, \sigma_t^{(m)}, \text{ENSO}_t)'$ for the extended model. The check function is defined as:

$$\rho_{\tau}(u) = u \cdot (\tau - \mathbf{1}_{u < 0}) = \begin{cases} \tau \cdot u & \text{if } u \geq 0, \\ (\tau - 1) \cdot u & \text{if } u < 0. \end{cases} \quad (9)$$

This asymmetric loss function penalizes under-prediction more heavily when τ is large and over-prediction more heavily when τ is small, ensuring that the estimated quantile achieves the target coverage rate. The minimization is a linear programming problem solved efficiently using the simplex algorithm (Koenker and d’Orey 1987). Standard errors are computed using the bootstrap.

3.3 Benchmark Models

We compare HAR-QR against three benchmarks that span the major modeling paradigms discussed in the literature review.

The first benchmark is *rolling historical volatility* with a one-year window ($w = 252$):

$$\hat{\sigma}_{t+h}^{\text{hist}} = \sqrt{\frac{1}{w} \sum_{i=1}^w (r_{t-i+1} - \bar{r})^2}. \quad (10)$$

For quantile forecasts, we compute empirical quantiles of the historical volatility distribution: sort the w past volatilities and select the value at position $\lceil \tau \cdot w \rceil$. This model captures only unconditional variation and serves as a naive baseline.

The second benchmark is the *GARCH(1,1)* model (Bollerslev 1986):

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (11)$$

For mean forecasts at horizon h , the GARCH(1,1) produces closed-form predictions:

$$\mathbb{E}_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2), \quad (12)$$

where $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ is the unconditional variance. For quantile forecasts, we simulate 10,000 paths assuming normal innovations and compute the τ -quantile of simulated average volatilities.

The third benchmark is *HAR-OLS*—the standard HAR model estimated by ordinary least squares, which provides forecasts of the conditional mean. Comparing HAR-OLS with HAR-QR at $\tau = 0.50$ tests whether the conditional distribution is symmetric; comparing HAR-QR at $\tau = 0.95$ with HAR-OLS tests whether tail forecasts add value beyond mean forecasts.

3.4 Evaluation Framework

Evaluating volatility forecasts is complicated by the fact that volatility is unobserved—every evaluation compares forecasts against an imperfect proxy (Patton 2011). We evaluate forecasts against Yang-Zhang volatility computed from realized prices, using loss functions that Patton (2011) has shown to be robust to noise in the volatility proxy.

For mean and median forecasts, we use QLIKE as the primary metric:

$$\text{QLIKE} = \frac{1}{T} \sum_{t=1}^T \left(\log \hat{\sigma}_t^2 + \frac{\sigma_t^2}{\hat{\sigma}_t^2} \right). \quad (13)$$

Patton (2011) derives necessary and sufficient conditions for the loss function to yield

forecast rankings that are robust to proxy noise, showing that among nine commonly used loss functions, only MSE and QLIKE satisfy these conditions. We prefer QLIKE because it penalizes under-prediction of volatility proportionally more than MSE, which is appropriate when under-prediction is costlier than over-prediction—a property particularly relevant for upper-quantile forecasts. As Puke and Schweikert (2026) demonstrate, aligning estimation and evaluation loss functions—what they term *coherent forecasting*—yields significant accuracy gains. Our HAR-QR models are coherent by construction: the check function (Equation 9) serves as both the estimation objective and the evaluation metric.

For quantile forecasts, we use quantile loss:

$$\text{QL}_\tau = \frac{1}{T} \sum_{t=1}^T \rho_\tau(\sigma_t - \hat{Q}_{\tau,t}). \quad (14)$$

To determine whether differences in forecast accuracy are statistically significant, we use the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the set of models $\mathcal{M}^*$ that contains the best model with a given confidence level, avoiding the need to specify a single benchmark. Starting from the full collection of competing models, the procedure iteratively tests for equal predictive ability and eliminates the worst-performing model until non-rejection occurs. We implement MCS with 10,000 bootstrap replications; models in $\mathcal{M}^*$ at the 90% level are considered statistically equivalent.

The forecast evaluation is conducted out-of-sample using a rolling estimation window of 2,520 trading days (approximately 10 years). At each forecast origin t , we estimate parameters using observations from $t - 2519$ to t , generate forecasts, and roll forward by one day. The window rolls *daily* to maximize out-of-sample forecasts for statistical evaluation. In practice, weekly or monthly re-estimation would be more efficient; the thesis uses daily rolling for evaluation purposes. With approximately 30 years of out-of-sample data, this yields thousands of forecast-realization pairs for statistical testing.

As a supplementary exercise, we freeze the estimation window at December 2021—before the 2022–2024 price crisis—and generate forecasts without updating parameters. This exercise does *not* test whether the model “predicted” the crisis. The 2022–2024 episode involved supply shocks (CSSVD virus, adverse weather) not directly observable in price data. Rather, the test asks whether the model’s structural relationships remain stable under stress. If pre-crisis parameters produce upper-quantile forecasts that cover realized volatility during the crisis, this demonstrates useful bounds even during extreme conditions.

Several assumptions underpin the methodology. First, we assume *local stationarity*: volatility dynamics are stable within each estimation window, though they may shift

across windows. The rolling-window design accommodates gradual parameter changes. Second, we assume that the Yang-Zhang estimator provides a reasonable proxy for latent volatility, which we validate empirically in Section 4. Third, we maintain *linear conditional quantiles* for parsimony, acknowledging that non-linear extensions could potentially improve forecasts. Fourth, we assume ENSO is exogenous to the cocoa market, which is physically reasonable since Pacific Ocean temperatures cannot be caused by cocoa prices.

Based on the literature, we expect that at short horizons (1 day to 1 month), HAR-QR will match or outperform benchmarks. At long horizons (6–12 months), standard HAR-QR will lose predictive power, but HAR-X-QR augmented with ENSO will recover predictive value—demonstrating that climate information provides incremental value precisely where the autoregressive signal fades.

4 Data

This section describes the data used to estimate and evaluate the volatility forecasting models developed in Section 3. Two primary datasets are required: daily futures prices for ICE London cocoa contracts and the Niño 3.4 climate index measuring El Niño–Southern Oscillation (ENSO) activity. After describing the cocoa futures data—including contract specifications, the construction of a continuous price series, and the variables derived from it—we turn to the ENSO index used as an exogenous predictor at long horizons. The section concludes with descriptive statistics documenting the stylized facts of cocoa returns and volatility.

4.1 Cocoa Futures Data

The primary data consist of daily Open-High-Low-Close (OHLC) prices and trading volume for ICE Futures Europe cocoa futures contracts. ICE London cocoa futures (contract code LCC) are standardized contracts for the delivery of 10 metric tonnes of cocoa beans. Each contract specifies delivery in one of five months: March, May, July, September, or December. Prices are quoted in British pounds sterling (GBP) per metric tonne, and trading occurs on the IntercontinentalExchange (ICE) electronic platform during European market hours. The data were obtained from Refinitiv Workspace via the corporate license held by Ritter Sport GmbH & Co. KG, ensuring institutional-grade data quality sourced directly from exchange feeds.

Continuous series construction. Futures contracts have finite maturity, requiring the construction of a continuous price series for volatility analysis. We use the generic front-month continuous contract, denoted LCCc1 in the Refinitiv system. This series

rolls to the next available contract approximately one month before expiration, maintaining exposure to the most liquid contract while avoiding delivery. The front-month contract typically exhibits the highest trading volume and the tightest bid-ask spreads, making it the standard choice for volatility research on commodity futures (Degiannakis et al. 2022; Alfeus and Nikitopoulos 2022). Rolling between contracts creates artificial price jumps when the new contract trades at a different level than the expiring one (the roll yield). For volatility estimation using OHLC data, these jumps are largely captured within the daily range and do not require explicit adjustment. The Yang-Zhang estimator (Section 3.1) is specifically designed to be robust to opening jumps, which include both overnight news and contract roll effects.

Sample period. The daily dataset spans 13,090 trading days from April 1, 1974, to December 15, 2025. This extended sample offers several advantages. It encompasses multiple El Niño and La Niña cycles, providing sufficient variation in the ENSO predictor for long-horizon models. It includes diverse market regimes: the 1977–1979 cocoa price spike driven by supply disruptions in West Africa, the long period of relative stability from 1990–2010, the financialization era beginning around 2008, and the recent 2022–2024 price surge attributed to the Cocoa Swollen Shoot Virus Disease (CSSVD) outbreak. The sample also includes approximately one year of high-frequency data (February 3, 2025, through January 12, 2026, comprising 20,791 five-minute price observations), which enables validation of the Yang-Zhang volatility estimator against realized volatility computed from intraday returns.

Table 2 summarizes the coverage and key characteristics of the cocoa futures data.

Table 2: Summary of Cocoa Futures Data

Characteristic	Value
Exchange	ICE Futures Europe
Contract	LCC (Cocoa Futures)
Contract size	10 metric tonnes
Price quotation	GBP per metric tonne
Continuous series	LCCc1 (front-month rolling)
Data source	Refinitiv Workspace
Sample period (daily)	April 1, 1974 – December 15, 2025
Number of observations (daily)	13,090 trading days
Sample period (5-minute)	February 3, 2025 – January 12, 2026
Number of observations (5-minute)	20,791 intraday observations
Variables (daily)	Open, High, Low, Close, Volume
Variables (5-minute)	Open, High, Low, Close, Volume, Bid, Ask

Variables and transformations. For each trading day t , the dataset provides five price observations: the opening price O_t , the highest price during the trading session H_t , the lowest price L_t , the closing price C_t , and total trading volume V_t . From these raw prices, we compute log returns:

$$r_t = \ln \left(\frac{C_t}{C_{t-1}} \right), \quad (15)$$

where C_t and C_{t-1} denote the closing prices on days t and $t-1$, respectively. Log returns are standard in financial econometrics due to their time-additivity and approximate normality over short horizons (Tsay 2010).

From the daily OHLC prices, we construct volatility estimates using the Yang-Zhang estimator detailed in Equation (1). The estimator combines overnight variance (close-to-open), intraday range-based variance (Rogers-Satchell), and close-to-close variance into a minimum-variance unbiased estimator that is drift-independent and robust to opening gaps (Yang and Zhang 2000). The Yang-Zhang estimator serves as the dependent variable in all volatility forecasting models and as the realized measure against which forecasts are evaluated.

For the one-year period with available intraday data, we compute realized volatility as the sum of squared five-minute log returns:

$$\text{RV}_t = \sum_{i=1}^M r_{t,i}^2, \quad (16)$$

where $r_{t,i}$ is the log return for the i -th five-minute interval on day t , and M is the number of five-minute intervals in the trading day (approximately 78 intervals for a 6.5-hour trading session). This realized volatility measure provides a benchmark for validating the Yang-Zhang estimator over the recent subsample.

Data quality. The daily cocoa futures data from Refinitiv Workspace contain no missing values for trading days. Non-trading days (weekends, exchange holidays) are excluded from the dataset, so the overnight return $o_t = \log(O_t/C_{t-1})$ entering the Yang-Zhang estimator spans the full gap between consecutive trading days—three calendar days over weekends, more over holidays. We do not rescale overnight returns by the number of intervening calendar days: the larger return after a weekend or holiday reflects genuine price risk that accumulated during the market closure, and including it without normalisation avoids additional distributional assumptions. The continuous front-month series (LCCc1) maintains coverage throughout the sample period, with automatic rolling between contracts ensuring no gaps.

4.2 ENSO Climate Data

As discussed in Section 2.5, ENSO provides a physically motivated predictor for cocoa volatility at long horizons. We measure ENSO activity using the Niño 3.4 index—sea surface temperature anomalies in the central equatorial Pacific (5°N–5°S, 170°W–120°W), computed as a three-month running mean relative to a climatological base period. Monthly values are obtained from the NOAA Climate Prediction Center, calculated using the ERSSTv5 dataset and publicly available at <https://psl.noaa.gov/data/correlation/nina34.data>.

Our sample covers January 1960 through July 2025, providing 786 monthly observations. NOAA defines El Niño (La Niña) episodes as periods when the index exceeds +0.5°C (falls below −0.5°C) for at least five consecutive months; the sample contains 15 El Niño and 13 La Niña episodes. Since the index is monthly while cocoa futures trade daily, each trading day inherits the Niño 3.4 value of its calendar month, as described in Section 3.2. The data contain no missing values through June 2025; placeholder entries (−99.99) for August 2025 onward do not affect the analysis because the out-of-sample period ends before these values are needed.

Table 3 summarizes the Niño 3.4 data.

Table 3: Summary of Niño 3.4 Climate Data

Characteristic	Value
Measures	Sea surface temperature anomaly
Region	Central equatorial Pacific (5°N–5°S, 170°W–120°W)
Data source	NOAA Climate Prediction Center
Dataset	ERSSTv5
URL	https://psl.noaa.gov/data/correlation/nina34.data
Sample period	January 1960 – July 2025
Observations	786 months
Frequency	Monthly
El Niño threshold	> +0.5°C (5 consecutive months)
La Niña threshold	< −0.5°C (5 consecutive months)

4.3 Descriptive Statistics and Stylized Facts

This subsection presents the empirical characteristics of cocoa returns and volatility, establishing the stylized facts that motivate the modeling choices in Section 3.

Price and return characteristics. Figure 1 displays the daily closing price of the front-month cocoa futures contract over the full sample period. The series exhibits several distinct episodes. The late 1970s witnessed a pronounced price spike, with cocoa reaching approximately 3,400 GBP per tonne in 1977–1979, driven by adverse weather

and supply disruptions in West Africa. Following this episode, prices stabilized at lower levels through the 1990s and 2000s, with moderate fluctuations around a mean of approximately 1,300 GBP per tonne. The 2010s showed gradual price increases, followed by the recent 2022–2024 surge reaching over 10,000 GBP per tonne—unprecedented in the modern trading history of cocoa—driven by the CSSVD outbreak in key producing regions.

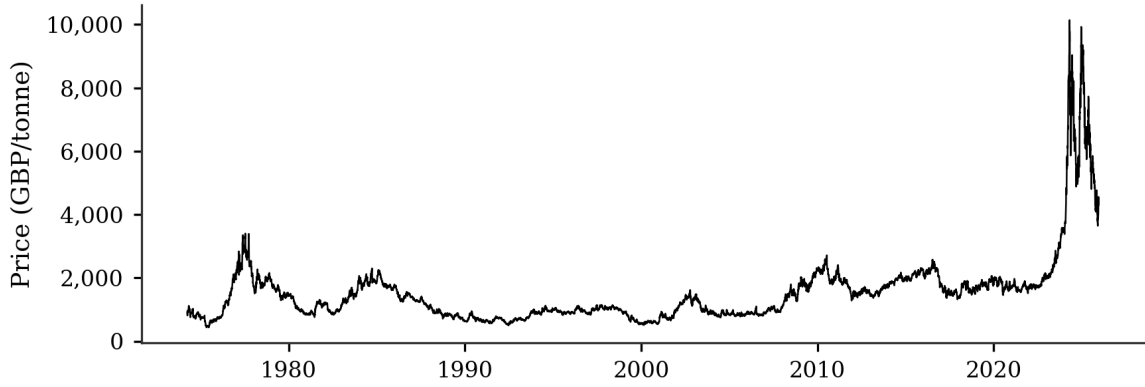


Figure 1: ICE London Cocoa Futures Daily Closing Price (1974–2025)

Notes: Front-month continuous contract (LCCc1) daily closing prices in GBP per metric tonne. Data source: Refinitiv Workspace. Sample period: April 1, 1974 – December 15, 2025 (13,090 trading days).

Table 4 presents descriptive statistics for daily log-returns. The mean daily return is slightly positive (0.012%), corresponding to an approximate annualized mean return of 3% over the full sample. However, returns exhibit substantial volatility: the standard deviation of 1.93% implies an annualized volatility of approximately 30%, consistent with cocoa’s classification as a high-volatility commodity. The distribution of returns deviates sharply from normality. Negative skewness (-0.997) indicates a left tail longer than the right, reflecting the tendency for large negative price movements to occur more frequently than large positive ones. Excess kurtosis of 12.96 documents fat tails: extreme returns occur far more often than a Gaussian distribution would predict. The Jarque-Bera test decisively rejects normality ($p < 0.001$), confirming that standard volatility models assuming Gaussian innovations may be misspecified for cocoa returns.

Volatility time series. Figure 2 displays the Yang-Zhang volatility estimator (annualized, in percent) computed from daily OHLC data. The figure reveals pronounced volatility clustering: periods of high volatility persist for extended intervals before reverting to lower levels. The late-1970s episode and the 2022–2024 crisis stand out as the two dominant volatility regimes in the sample. Between these episodes, volatility fluctuates around a lower baseline with occasional spikes corresponding to supply shocks, weather events, and financial market turbulence (e.g., the 2008 financial crisis).

Table 4: Descriptive Statistics for Daily Log-Returns

Statistic	Value
Observations	13,087
Mean (%)	0.012
Median (%)	0.000
Std. Deviation (%)	1.929
Minimum (%)	−26.212
Maximum (%)	12.988
25th Percentile (%)	−0.917
75th Percentile (%)	0.945
Skewness	−0.997
Excess Kurtosis	12.963
Jarque-Bera	141,013
<i>p</i> -value	< 0.001

Notes: Daily log-returns computed as $r_t = \ln(C_t/C_{t-1})$ for ICE London cocoa front-month continuous contract (LCCc1), April 1974–December 2025. Returns expressed in percentage points. Excess kurtosis reported (normal distribution: 0). Jarque-Bera test strongly rejects normality ($p < 0.001$), consistent with fat tails and negative skewness typical of commodity returns.

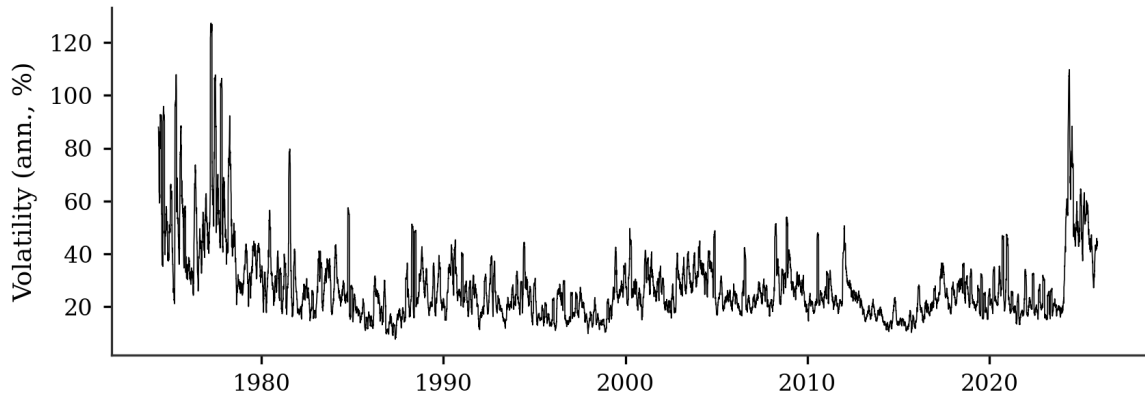


Figure 2: Yang-Zhang Annualized Volatility Time Series (1974–2025)

Notes: Yang-Zhang volatility estimator computed from daily OHLC prices and annualized (percent). The estimator combines overnight, intraday, and close-to-close variance components following Yang and Zhang (2000). Pronounced volatility clustering and regime-switching are evident, motivating the use of HAR models that capture persistence across multiple horizons.

The persistence of volatility justifies the HAR model’s multi-horizon structure: current volatility depends not only on yesterday’s volatility (daily component) but also on average volatility over the past week and month (weekly and monthly components). The clustering pattern also motivates quantile regression: during high-volatility regimes, the tails of the volatility distribution shift outward, and point forecasts of the mean may underestimate the risk of extreme outcomes.

Stationarity. The regression models in Section 3 require stationary inputs. Table 5 reports Augmented Dickey–Fuller (ADF) and Phillips–Perron (PP) unit root tests for the three key series. Both tests include a constant; ADF lag length is selected by AIC.

Table 5: Unit root tests for stationarity

Series	N	ADF		PP	
		Statistic	p -value	Statistic	p -value
Log returns	13,087	−112.08	< 0.001	−112.09	< 0.001
Yang-Zhang volatility	13,087	−8.42	< 0.001	−90.99	< 0.001
Niño 3.4 anomaly	787	−6.85	< 0.001	−6.46	< 0.001

Note: ADF = Augmented Dickey–Fuller test (Dickey and Fuller 1979); PP = Phillips–Perron test (Phillips and Perron 1988). Both tests include a constant. ADF lag length selected by AIC. The null hypothesis is a unit root; rejection indicates stationarity.

All three series decisively reject the unit root null at the 1% level. Log returns are stationary by construction, with ADF and PP statistics both below −112. The Yang-Zhang volatility series exhibits high persistence—the ADF test requires 39 lags to whiten the residuals—but remains stationary (ADF = −8.42, $p < 0.001$). The large discrepancy between the ADF and PP statistics (−8.42 vs. −90.99) reflects the difference in how each test handles serial correlation: ADF augments the regression with lagged differences, while PP applies a non-parametric correction to the test statistic. Both agree on rejection. The Niño 3.4 anomaly is stationary by design, as it measures deviations from a climatological mean, confirmed by both tests (ADF = −6.85, PP = −6.46, $p < 0.001$). These results justify the use of standard OLS and quantile regression methods without differencing or cointegration corrections.

The data described in this section provide the foundation for the empirical analysis. The extended daily sample (1974–2025) ensures sufficient variation in both volatility regimes and ENSO phases, while the availability of intraday data for the recent period enables validation of the Yang-Zhang volatility proxy. The next section presents the empirical results from the HAR-based quantile regression models estimated using these data.

5 Empirical Results

This section presents the out-of-sample forecasting results, organized around the two research questions. Section 5.1 compares mean forecast accuracy across the three baseline models. Section 5.2 evaluates distributional forecasts at the upper tail, testing whether HAR-QR outperforms the benchmarks across horizons (RQ1). Section 5.3 assesses whether the Niño 3.4 climate index adds forecast value at long horizons (RQ2). Section 5.4 tests the statistical significance of the observed differences using Model Confidence Sets. Section 5.5 synthesizes the findings. All results use a rolling estimation window of 2,520 trading days with daily re-estimation, yielding approximately 10,500 out-of-sample forecast–realization pairs at each horizon.

5.1 Mean Forecast Comparison

Table 6 reports the QLIKE loss for the three mean forecast models across all six horizons. Figure 3 visualizes the same results. The ranking is unambiguous: HAR-OLS achieves the lowest (best) QLIKE at every horizon, followed by GARCH(1,1), with historical volatility uniformly last.

Table 6: Mean forecast QLIKE loss by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	−7.002	−7.256	−7.327	−7.334*	−7.324	−7.310
GARCH(1,1)	−7.242	−7.388	−7.406*	−7.381	−7.356	−7.345
HAR-OLS	−7.302	−7.482	−7.521*	−7.515	−7.492	−7.485

Note: QLIKE loss (Patton 2011); more negative values indicate better forecast performance. Bold: best model at each horizon. *Best horizon for each model. Evaluation period: approximately 10,500 out-of-sample trading days (1984–2025).

The gap between HAR-OLS and GARCH is largest at the one-month horizon (0.115 QLIKE units) and narrowest at one day (0.060 units). This pattern reflects the structural differences between the two models. GARCH(1,1) captures short-run volatility dynamics effectively—its conditional variance process adapts quickly to recent shocks. But its exponential memory decay means that multi-step forecasts converge rapidly toward the unconditional variance, erasing the model’s informational advantage at longer horizons. HAR-OLS, by contrast, includes a monthly volatility component ($\sigma_t^{(m)}$) that captures persistence at lower frequencies. This component maintains a correlation of 0.66 with the twelve-month target, compared to 0.37 for the daily component, explaining why HAR-OLS retains a substantial edge even at twelve months.

Historical volatility performs worst because it is the most constrained: its trailing average cannot distinguish between a volatility spike driven by a transient shock and a persistent regime change. Nevertheless, all three models share a common hori-

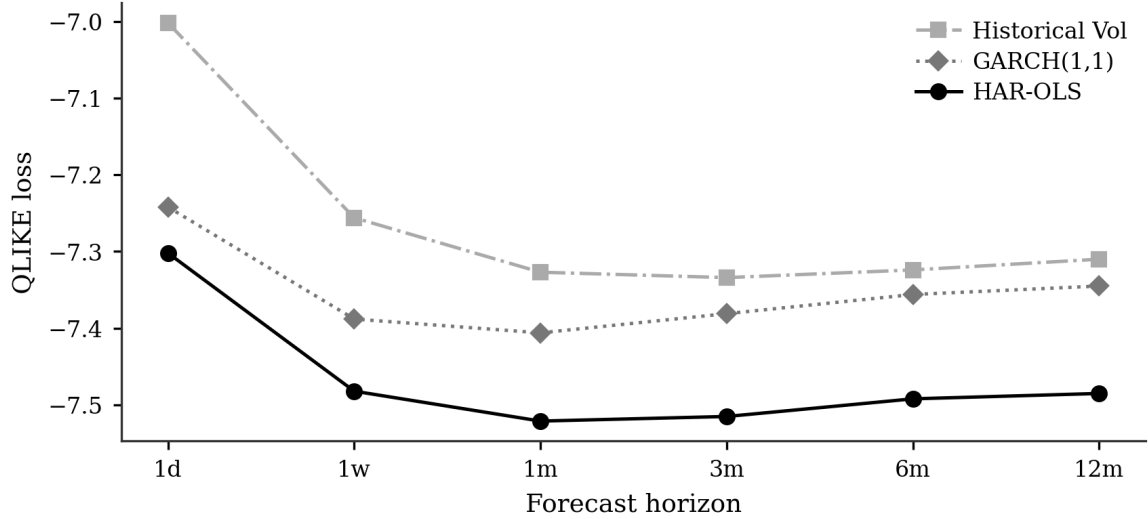


Figure 3: QLIKE loss across forecast horizons by model

Notes: More negative values indicate better mean forecast accuracy. HAR-OLS dominates both benchmarks at every horizon; the gap widens from one day to one month before stabilizing.

zon pattern: loss improves from the one-day to the one-month horizon—reflecting the smoothing effect of averaging in the target $\bar{\sigma}_{t:t+h}$ —before degrading at six and twelve months as the autoregressive signal weakens. The fact that even HAR-OLS degrades beyond one month confirms the fundamental difficulty of long-horizon volatility forecasting and motivates the search for additional predictive signals in Section 5.3.

5.2 Quantile Forecast Comparison (RQ1)

Mean forecasts summarize only the center of the conditional distribution. For risk assessment, the upper tail matters: a model that accurately forecasts average volatility may still understate the probability of extreme outcomes. We now compare four models on their ability to forecast the 95th percentile of future volatility. Historical volatility uses the empirical quantile of the trailing 252-day distribution. GARCH(1,1) derives quantiles from 5,000 simulated multi-step return paths. HAR-QR estimates the quantile directly through quantile regression on the HAR structure, and HAR-X-QR augments this with the Niño 3.4 index.

Table 7 reports the quantile loss at $\tau = 0.95$ for all four models. Figure 4 presents the same comparison graphically.

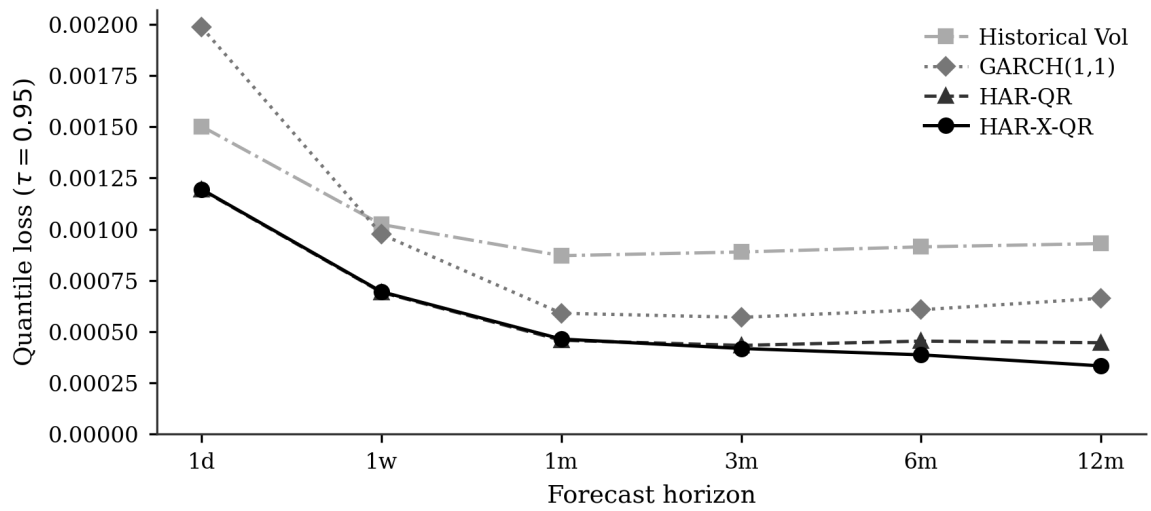
Three patterns emerge from this comparison.

First, HAR-QR dominates both benchmarks at every horizon. The quantile loss of HAR-QR is 21% below historical volatility and 40% below GARCH at the one-day horizon; 47% below historical volatility and 22% below GARCH at one month. This consistent advantage stems from quantile regression’s ability to model the conditional

Table 7: Upper-tail quantile loss ($\tau = 0.95$) by model and horizon

Model	1d	1w	1m	3m	6m	12m
Historical Vol	0.001502	0.001023	0.000871*	0.000889	0.000914	0.000930
GARCH(1,1)	0.001986	0.000975	0.000590	0.000570*	0.000607	0.000663
HAR-QR	0.001193	0.000692	0.000458	0.000433*	0.000454	0.000446
HAR-X-QR	0.001195	0.000695	0.000464	0.000418	0.000387	0.000333*

Note: Quantile loss as defined in Equation (14). Lower values indicate better forecasts. Bold: best model at each horizon. *Best horizon for each model. Historical Vol quantiles are empirical percentiles of the trailing 252-day volatility distribution. GARCH quantiles are derived from 5,000 simulated multi-step paths.

Figure 4: Upper-tail quantile loss ($\tau = 0.95$) across forecast horizons

Notes: Lower values indicate better upper-tail forecasts. HAR-QR dominates both benchmarks at all horizons. HAR-X-QR (ENSO-augmented) separates from HAR-QR at horizons beyond three months. The y-axis starts at zero for honest visual comparison.

quantile directly, rather than deriving it from a fitted mean or variance. HAR-QR conditions on the same heterogeneous volatility components as HAR-OLS but targets the 95th percentile, capturing tail-specific dynamics that neither the empirical quantile (historical volatility) nor the parametric simulation (GARCH) can replicate.

Second, the relative performance of the benchmarks reverses across horizons. At the one-day horizon, GARCH(1,1) produces the worst upper-tail forecast of all four models (0.001986), even exceeding the simple historical benchmark (0.001502). This occurs because GARCH’s one-step-ahead simulation amplifies model misspecification in the tails: the assumed normal distribution of standardized returns underestimates kurtosis, causing the simulated 95th percentile to be systematically misplaced. From one week onward, GARCH improves rapidly and outperforms historical volatility at all remaining horizons. The reversal reflects GARCH’s core trade-off: its parametric structure adds value when it correctly captures the variance dynamics (short-to-medium horizons), but at longer horizons its mean-reversion property causes the simulated distribution to narrow toward the unconditional one, limiting the quality of its tail forecasts. By twelve months, GARCH’s quantile loss (0.000663) is still 49% higher than HAR-QR’s (0.000446).

Third, HAR-X-QR separates from HAR-QR only at long horizons. At one day through one month, the two HAR models are virtually indistinguishable (differences $< 3\%$). The divergence begins at three months (+3.26% improvement) and accelerates to +14.66% at six months and +25.44% at twelve months. This horizon dependence is analyzed in detail in Section 5.3.

These findings answer RQ1: the HAR-QR model forecasts the upper tail of the conditional volatility distribution more accurately than both the historical volatility benchmark and GARCH(1,1) at all horizons from one day to twelve months.

Figures 5 and 6 illustrate this comparison visually. Each figure overlays the $\tau = 0.95$ forecast boundary from all four models on realized volatility for the 2015–2025 window, which includes the 2022–2024 cocoa supply crisis.

At the one-month horizon (Figure 5), the HAR-QR and HAR-X-QR boundaries track the realized upper tail closely, rising in tandem with the crisis spike. GARCH’s boundary reacts with a visible lag, and historical volatility’s boundary is the slowest to adjust. At the twelve-month horizon (Figure 6), the differences are more pronounced: the HAR-X-QR boundary rises ahead of the crisis, reflecting the ENSO signal’s advance warning of supply stress, while the GARCH boundary remains comparatively flat—a direct consequence of its mean-reverting multi-step forecasts converging toward the unconditional variance.

Table 8 reports violation rates for the HAR-QR upper quantile ($\tau = 0.95$) by horizon. Under correct conditional coverage, the violation rate should equal $1 - \tau = 5\%$. At the one-day and one-week horizons, the model achieves near-target coverage (4.84%

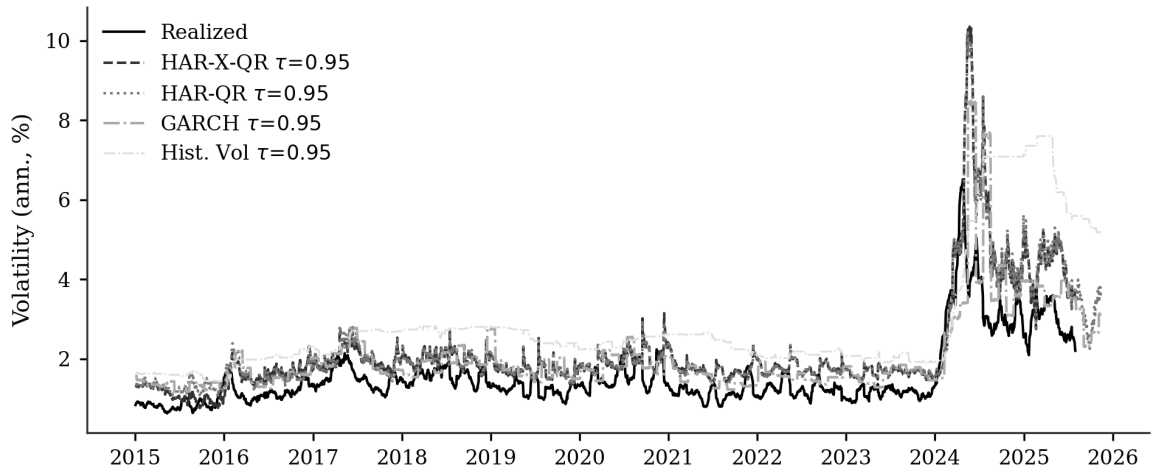


Figure 5: Upper-tail ($\tau = 0.95$) forecast comparison at the one-month horizon

Notes: Realized Yang-Zhang volatility (solid black) and $\tau = 0.95$ quantile forecasts from four models. Zoomed to 2015–2025 to show the 2022–2024 supply crisis period in detail.

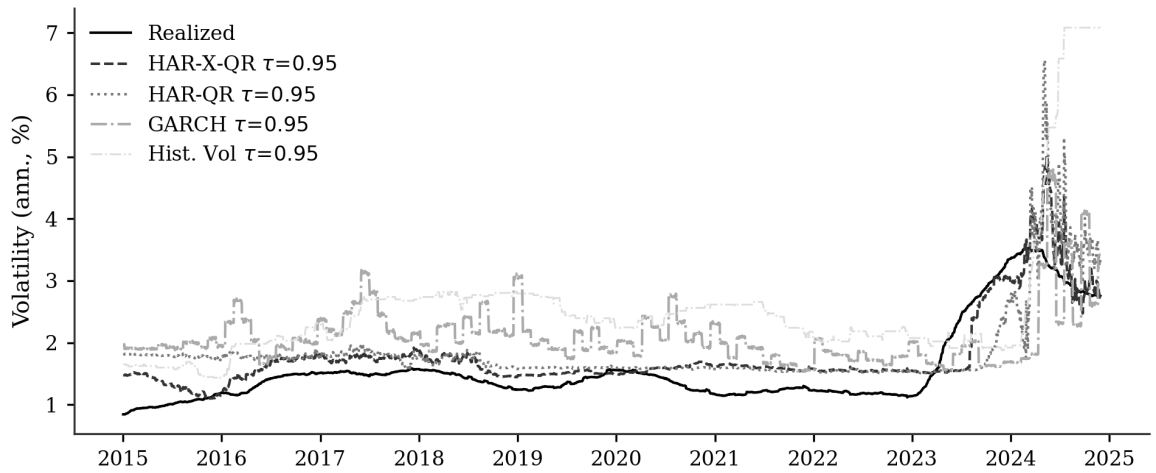


Figure 6: Upper-tail ($\tau = 0.95$) forecast comparison at the twelve-month horizon

Notes: Same construction as Figure 5. At the twelve-month horizon, HAR-X-QR's boundary rises ahead of the crisis, reflecting the ENSO signal, while GARCH's boundary remains comparatively flat due to mean-reversion in multi-step simulations.

and 5.15%). Coverage remains acceptable through the three-month horizon (5.90%), but exceeds the target at six months (7.04%) and twelve months (8.34%). The coverage degradation at long horizons indicates that the autoregressive signal alone becomes insufficient to track extreme upper-tail volatility—the HAR structure, while dominant among the models tested, still underestimates how far the distribution can shift in periods of sustained supply stress. This motivates the ENSO augmentation examined next.

Table 8: HAR-QR upper-quantile ($\tau = 0.95$) coverage by horizon

Horizon	N	Violations	Violation rate (%)
1d	10,545	510	4.84
1w	10,541	543	5.15
1m	10,524	600	5.70
3m	10,480	618	5.90
6m	10,414	733	7.04
12m	10,282	857	8.34

Note: A violation occurs when realized volatility exceeds the $\tau = 0.95$ quantile forecast. Under correct specification, the violation rate should equal 5%. N denotes the number of forecast–realization pairs at each horizon.

5.3 ENSO Augmentation: HAR-X-QR (RQ2)

Table 9 reports the percentage change in quantile loss when moving from HAR-QR to HAR-X-QR across all five quantile levels and six horizons. Positive values indicate improvement (lower loss under HAR-X-QR).

Table 9: Quantile loss improvement (%) from ENSO augmentation (HAR-X-QR vs. HAR-QR)

τ	1d	1w	1m	3m	6m	12m
0.05	−0.1081	−0.2296	−0.4086	−0.3029	−0.3521	+4.0532
0.25	+0.0378	−0.0273	−0.3798	−0.9790	+0.2137	+3.5822
0.50	−0.0076	−0.2093	−0.2698	−0.3994	−0.0643	+1.1225
0.75	−0.0357	−0.1810	−0.6684	−1.3689	−0.5717	−0.8660
0.95	−0.1628	−0.5111	−1.4628	+3.2564	+14.6602	+25.4375

Note: Positive values indicate that HAR-X-QR achieves lower quantile loss than HAR-QR. Computed as $(1 - \text{QL}_{\text{HAR-X}}/\text{QL}_{\text{HAR}}) \times 100$.

The results reveal a striking horizon- and quantile-dependent pattern that aligns with the physical transmission mechanism of ENSO to cocoa markets. At short horizons (one day to one month), ENSO augmentation provides no benefit and occasionally worsens forecasts by small margins. This is expected: ocean surface temperature

anomalies affect cocoa production through the growing season, a process that operates over months to years. At such short horizons, the Niño 3.4 coefficient captures noise rather than signal, and the additional parameter slightly degrades out-of-sample performance.

The first meaningful improvement appears at three months for the upper tail: $\tau = 0.95$ quantile loss decreases by 3.26%. The effect strengthens substantially at six months (+14.66%) and reaches its maximum at twelve months (+25.44%). This concentration in the upper tail is economically interpretable. El Niño conditions increase the probability of drought-related supply disruptions in West Africa, which manifest as spikes in price volatility—rightward shifts in the upper tail—rather than a uniform upward shift across the entire distribution. The GARCH and historical volatility benchmarks cannot exploit this climate signal, explaining why their twelve-month upper-tail performance in Table 7 falls far behind HAR-X-QR: historical volatility’s loss (0.000930) is 179% higher than HAR-X-QR’s (0.000333), and GARCH’s (0.000663) is 99% higher.

At twelve months, improvement also appears at the lower quantiles (+4.05% at $\tau = 0.05$, +3.58% at $\tau = 0.25$), suggesting that at sufficiently long horizons, ENSO carries information about the general shape of the conditional distribution, not only its upper extreme. The median ($\tau = 0.50$) shows only modest improvement (+1.12%), confirming that the benefit of ENSO augmentation is predominantly distributional rather than centered on the mean.

Table 10 examines how ENSO augmentation affects coverage calibration. For the lower tail ($\tau = 0.05$), HAR-X-QR consistently improves coverage at long horizons: at twelve months, the violation rate falls from 10.69% under HAR-QR to 9.28% under HAR-X-QR, moving closer to the nominal 5%. For the upper tail ($\tau = 0.95$), the picture is more nuanced. HAR-X-QR achieves a substantially lower quantile loss (Table 7) but its violation rate at twelve months rises to 10.29%, compared to 8.34% under HAR-QR. This apparent paradox reflects the model’s recalibration of the upper boundary: HAR-X-QR positions the $\tau = 0.95$ quantile more aggressively in El Niño periods, anticipating large volatility spikes. When these spikes do not fully materialize, the boundary is breached. The lower quantile loss indicates that, on average across all out-of-sample days, the HAR-X-QR boundary is better placed; the higher violation rate indicates that it is breached more often in specific episodes. This trade-off is consistent with a model that correctly identifies the direction of climate risk but cannot perfectly time its realization.

Table 10: Coverage comparison: violation rates (%) for HAR-QR and HAR-X-QR

Horizon	Lower tail ($\tau = 0.05$)		Upper tail ($\tau = 0.95$)	
	HAR-QR	HAR-X-QR	HAR-QR	HAR-X-QR
1d	4.77	4.74	4.82	4.80
1w	5.23	5.16	5.19	5.22
1m	6.76	6.48	5.68	6.13
3m	7.89	7.39	5.91	6.49
6m	9.59	8.61	7.04	8.91
12m	10.69	9.28	8.34	10.29

Note: Under correct specification, both violation rates should equal 5%. Values closer to 5% indicate better calibration. $N \approx 10,400$ – $10,500$ forecast–realization pairs per horizon.

5.4 Statistical Significance: Model Confidence Sets

The preceding sections compare models by point estimates of loss functions. To assess whether the observed differences are statistically significant, we apply the Model Confidence Set (MCS) procedure of Hansen et al. (2011). The MCS identifies the set $\mathcal{M}^*$ of models whose forecasting ability cannot be distinguished from that of the best model at a given confidence level. We use 10,000 stationary bootstrap replications and the max- t statistic, constructing $\mathcal{M}^*$ at the 90% confidence level.

Table 11 reports MCS results for QLIKE loss. HAR-OLS is the sole member of $\mathcal{M}^*$ at the one-day and one-week horizons ($p = 1.00$), confirming that its mean forecast superiority over GARCH and historical volatility is statistically significant at short horizons. From one month onward, the HAR-QR and HAR-X-QR median forecasts enter $\mathcal{M}^*$, indicating that their mean forecast accuracy is statistically indistinguishable from HAR-OLS at longer horizons. GARCH(1,1) is excluded from $\mathcal{M}^*$ at every horizon—its inferior QLIKE performance is not attributable to sampling variation.

Table 12 presents MCS results for quantile loss at $\tau = 0.95$, the central evaluation criterion for distributional forecasting. The results are unambiguous: at every horizon, only the HAR-based quantile regression models belong to $\mathcal{M}^*$. Historical volatility and GARCH(1,1) are excluded at all six horizons, with GARCH achieving $p < 0.01$ at one day through one month. This confirms that the HAR-QR advantage documented in Table 7 is statistically significant, not merely numerical.

Within the HAR family, a revealing pattern emerges. At the one-day horizon, HAR-QR is the sole member of $\mathcal{M}^*$ ($p = 1.00$), with HAR-X-QR narrowly excluded ($p = 0.05$). From one week onward, both models coexist in $\mathcal{M}^*$. The critical shift occurs at three months: HAR-X-QR takes $p = 1.00$ from HAR-QR, indicating that the ENSO-augmented model becomes the top-ranked specification. This reversal persists at six and twelve months ($p = 1.00$ for HAR-X-QR versus $p = 0.18$ and $p = 0.19$ for HAR-QR). While the two models remain statistically indistinguishable at the 90%

Table 11: Model Confidence Set: QLIKE loss (mean forecasts)

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.01	0.04	0.07	0.11	0.10	0.06
	$\mathcal{M}^*$				✓	✓	
GARCH(1,1)	p	0.01	0.00	0.00	0.00	0.00	0.01
	$\mathcal{M}^*$						
HAR-OLS	p	1.00	1.00	1.00	1.00	1.00	1.00
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-QR (med.)	p	0.00	0.04	0.10	0.39	0.30	0.24
	$\mathcal{M}^*$			✓	✓	✓	✓
HAR-X-QR (med.)	p	0.00	0.04	0.10	0.31	0.30	0.24
	$\mathcal{M}^*$			✓	✓	✓	✓

Note: MCS p -values from Hansen et al. (2011); max- t statistic, 10,000 stationary bootstrap replications, 90% confidence level. ✓ indicates membership in $\mathcal{M}^*$. Bold: $p = 1.00$ (best model).

HAR-QR and HAR-X-QR evaluated on their median ($\tau = 0.50$) forecasts.

level, the consistent shift in ranking from three months onward provides supporting evidence for RQ2: the Niño 3.4 index adds genuine predictive content at long horizons.

Table 12: Model Confidence Set: quantile loss at $\tau = 0.95$

Model		Horizon					
		1d	1w	1m	3m	6m	12m
Historical Vol	p	0.02	0.00	0.00	0.00	0.00	0.00
	$\mathcal{M}^*$						
GARCH(1,1)	p	0.00	0.00	0.02	0.04	0.03	0.05
	$\mathcal{M}^*$						
HAR-QR	p	1.00	1.00	1.00	0.51	0.18	0.19
	$\mathcal{M}^*$	✓	✓	✓	✓	✓	✓
HAR-X-QR	p	0.05	0.20	0.36	1.00	1.00	1.00
	$\mathcal{M}^*$		✓	✓	✓	✓	✓

Note: Same MCS configuration as Table 11. ✓ indicates membership in $\mathcal{M}^*$ at the 90% confidence level. Bold: $p = 1.00$ (best model). The shift of $p = 1.00$ from HAR-QR to HAR-X-QR at three months reflects the onset of ENSO predictive power.

5.5 Summary of Findings

Table 13 consolidates the key metrics for all models across the horizon spectrum. Three conclusions emerge.

First, regarding RQ1, HAR-OLS achieves the lowest QLIKE at all horizons; HAR-QR achieves the lowest upper-tail quantile loss at horizons up to one month and remains

Table 13: Consolidated model comparison across horizons

Metric	Model	Horizon				
		1d	1m	3m	6m	12m
QLIKE	Historical Vol	−7.002	−7.327	−7.334*	−7.324	−7.310
	GARCH(1,1)	−7.242	−7.406*	−7.381	−7.356	−7.345
	HAR-OLS	−7.302	−7.521*	−7.515	−7.492	−7.485
QL ($\tau=0.95$)	Historical Vol	0.001502	0.000871*	0.000889	0.000914	0.000930
	GARCH(1,1)	0.001986	0.000590	0.000570*	0.000607	0.000663
	HAR-QR	0.001193	0.000458	0.000433*	0.000454	0.000446
	HAR-X-QR	0.001195	0.000464	0.000418	0.000387	0.000333*

Note: Bold: best model at each horizon. *Best horizon for each model. QLIKE: more negative is better. QL: lower is better. The 1w horizon is omitted for compactness.

within 4% of the best model beyond that. The MCS confirms statistical significance: GARCH(1,1) is excluded from $\mathcal{M}^*$ at every horizon for both loss criteria (Tables 11 and 12).

Second, regarding RQ2, the Niño 3.4 index provides substantial incremental value exclusively at long horizons and in the upper tail. At twelve months, the $\tau = 0.95$ quantile loss falls by 25.44% when moving from HAR-QR to HAR-X-QR—the single largest improvement in the study. The MCS p -value for HAR-X-QR shifts from 0.05 at one day to 1.00 at three months and beyond (Table 12).

Third, coverage analysis reveals a systematic challenge at long horizons. All models produce violation rates above the nominal 5% at six and twelve months, indicating that the conditional distribution of cocoa volatility is harder to characterize at procurement-relevant horizons than at short ones. ENSO augmentation improves lower-tail coverage but introduces a recalibration effect at the upper tail. These limitations are discussed further in Section ??.

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